

Case Study Assessment

Constructing and Analysing a Multi-Tier Supply Chain Network: The Case of CATL

Module: ESI4322 Supply Chain Engineering | Weighting: 35 % | Deadline: Nov 16 and 18 2026

1. Overview

Contemporary Amperex Technology Co., Ltd. (CATL) is the world's largest manufacturer of lithium-ion batteries, supplying electric-vehicle, energy-storage and consumer-electronics customers globally. Its supply base spans raw-material processors, component manufacturers, and assembly partners across multiple tiers, and its finished products flow onward to a wide range of industrial customers.

In this assessment you will use real, trade-record-derived data to construct a directed, multi-tier supply chain network centred on CATL, analyse its structure, and identify companies that occupy more than one tier of the network, a phenomenon with direct implications for how supply chain risk and dependency should be assessed.

2. Learning Outcomes

On successful completion of this assessment, you will be able to:

- Summarise a multi-tier supply chain's structure from tabular data using Python (pandas).
- Construct and visualise a directed supply chain network using networkx and matplotlib.
- Identify suppliers that span more than one tier of a supply chain.
- Interpret what a network diagram and a multi-tier supplier list reveal about supply chain structure.

3. Datasets Provided

You are provided with four CSV files. Company names are drawn from real trade records.

File	Represents	Columns you will use
Supplier1.csv	Tier 1 suppliers of CATL	Supplier1 Name — one row per Tier 1 supplier
Supplier2.csv	Tier 2 suppliers (suppliers to Tier 1)	Supplier2 Name — one row per Tier 2 supplier
Supplier3.csv	Tier 3 suppliers (suppliers to Tier 2)	Supplier3 Name — one row per Tier 3 supplier
Network_Edges.csv	Every shipment relationship (edge) in the network	Two columns giving each edge's source node and target node (e.g. a supplier name, or “CATL”)
Customers.csv	Tier 1 customers of CATL	Customer Name — one row per Tier 1 customer

4. Tasks

Task 1 — Tier counts and network edges (20 marks)

1. Load Supplier1.csv, Supplier2.csv and Supplier3.csv into Python and identify the number of unique Tier 1 suppliers, Tier 2 suppliers, and Tier 3 suppliers, and the total number of network edges.
2. Load Customers.csv, and identify the number of unique CATL's Tier 1 customers.

Task 2 — Constructing and visualising the network (20 marks)

3. Build a network graph (e.g. with networkx's Graph) from the Network_Edges.csv dataset, and report the total number of nodes and edges in the constructed graph.
4. Produce ONE diagram of the network using matplotlib (or an equivalent library), illustrating the relationships between CATL and the supplier tiers and customers.

Task 3 — Identifying suppliers in multiple tiers (20 marks)

A supplier may appear in more than one tier of CATL's network, for example, a company might be listed in both Supplier1 and Supplier2 (Tier 1 and Tier 2), in both Supplier2 and Supplier3 (Tier 2 and Tier 3), or in both Supplier1 and Supplier3 (Tier 1 and Tier 3), or across three tiers (Tiers 1, 2, and 3).

5. Identify multi-tier suppliers, report the total count together with the count for each of the three combinations above.
6. Produce a table (CSV or DataFrame) listing every multi-tier supplier showing whether it belongs to Tier 1, Tier 2 and/or Tier 3.

7. Guidance and Suggested Tools

You are free to use any Python libraries you are comfortable with. The following are suggested:

- pandas — loading the CSV files and converting name columns to sets.
- networkx — constructing the directed graph (DiGraph) from your edge list.
- matplotlib — producing your Task 2 diagram; a spring layout (nx.spring_layout) generally gives a readable overall shape for a network this size.

Start with a small, manually checked sample before running your code on the full dataset, and confirm your Task 1 counts against the sanity-check figures before moving on to Tasks 2 and 3, an error in Task 1 will propagate into both.

Please email your Python code to ying.xie@ucf.edu by 12:00pm on November 16, 2026 (code = 50% of the total grade: Task 1 = 30%, Task 2 = 30%, Task 3 = 40%).

Your team will then present and discuss your results on November 18, 2026, 9:00–10:00am, in CB1 212: 15 minutes for the presentation plus 5 minutes for Q&A (presentation = 50% of the total grade, weighted the same way: Task 1 = 30%, Task 2 = 30%, Task 3 = 40%).

8. Academic Integrity

This is an individual assessment. You may discuss general concepts with peers, but all code, analysis, and written interpretation submitted must be your own work. Cite any libraries, or generative-AI assistance you use, in line with UCF's academic integrity policy reference.